JINTAO ZHANG

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EDUCATION

Tsinghua University | Ph.D. in Computer Science and Technology — **Aug. 2024 – Now**
Supervisor: Prof. Jun Zhu Homepage, Prof. Jianfei Chen Homepage. — **Beijing, China**

Tsinghua University | Master in Computer Science and Technology — **Aug. 2021 – Jun. 2024**
Supervisor: Prof. Guoliang Li Homepage. — **Beijing, China**

Xidian University — **Sep. 2017 – Jun. 2021**
Bachelor of Science in Computer Science and Technology — **Shaanxi, China**

SELECTED PUBLICATIONS

[ICLR 2025] SageAttention: Accurate 8-Bit Attention for Plug-and-play Inference Acceleration [Code (4.1K Stars)]
Full Paper, CCF-A **Jintao Zhang**, Jia Wei, Pengle Zhang, Jun Zhu, Jianfei Chen.

[ICML 2025] SageAttention2: Efficient Attention with Thorough Outlier Smoothing and Per-thread INT4 Quantization
Full Paper, CCF-A **Jintao Zhang**, Haofeng Huang, Pengle Zhang, Jia Wei, Jun Zhu, Jianfei Chen. [Code (4.1K Stars)]

[ICLR 2026] SLA: Beyond Sparsity in Diffusion Transformers via Fine-Tunable Sparse-Linear Attention [Code (0.7K Stars)]
Full Paper, CCF-A **Jintao Zhang**, Haoxu Wang, Kai Jiang, ..., Ion Stoica, Joseph E. Gonzalez, Jianfei Chen, Jun Zhu.

[ICML 2025] SpargeAttention: Accurate Sparse Attention Accelerating Any Model Inference [Code (1.4K Stars)]
Full Paper, CCF-A **Jintao Zhang**, Chendong Xiang, Haofeng Huang, Haocheng Xi, Jia Wei, Jun Zhu, Jianfei Chen.

[Preprint] Vidu S1: A Real-Time Interactive Video Generation Model [Code (0.6K Stars)]
Technical Report **Jintao Zhang**, Kai Jiang, Jintao Chen, Xu Wang, ..., Zhijie Deng, Fan Bao, Jianfei Chen, Jun Zhu.

[Preprint] Vidu S2: Real-Time Interactive, Editable, and Spatial Video Generation [Code (0.6K Stars)]
Technical Report **Jintao Zhang**, Kai Jiang, Jintao Chen, Xu Wang, ..., Zhijie Deng, Fan Bao, Jianfei Chen, Jun Zhu.

[Preprint] TurboDiffusion: Accelerating Video Diffusion Models by 100-200 Times [Code (4.0K Stars)]
Technical Report **Jintao Zhang**, Kaiwen Zheng, Kai Jiang, Haoxu Wang, Ion Stoica, Joseph E. Gonzalez, Jianfei Chen, Jun Zhu.

[NeurIPS 2025] SageAttention3: Microscaling FP4 Attention for Inference and An Exploration of 8-Bit Training
Spotlight, CCF-A **Jintao Zhang**, Jia Wei, Pengle Zhang, ..., Haoxu Wang, Kai Jiang, Jianfei Chen, Jun Zhu. [Code (4.1K Stars)]

[NeurIPS 2026] SLA2: Sparse-Linear Attention with Learnable Routing and QAT [Code (0.7K Stars)]
Full Paper, CCF-A **Jintao Zhang**, Haoxu Wang, Kai Jiang, ..., Youhe Jiang, Ion Stoica, Jianfei Chen, Jun Zhu, Joseph E. Gonzalez.

[NeurIPS 2026] SpargeAttention2: Trainable Sparse Attention via Hybrid Top-k+Top-p Masking and Distillation Fine-Tuning
Full Paper, CCF-A **Jintao Zhang**, Kai Jiang, Chendong Xiang, ..., Yuezhou Hu, Haocheng Xi, Jianfei Chen, Jun Zhu. [Code (1.4K Stars)]

[ICDE 2025] SAGE: A Framework of Precise Retrieval for RAG
Full Paper, CCF-A **Jintao Zhang**, Guoliang Li, Jinyang Su.

[SIGMOD 2024] PACE: Poisoning Attacks on Learned Cardinality Estimation
Full Paper, CCF-A **Jintao Zhang**, Guoliang Li, Chao Zhang, Chengliang Chai.

[ICDE 2023] AutoCE: An Accurate and Efficient Model Advisor for Learned Cardinality Estimation
Full Paper, CCF-A **Jintao Zhang**, Chao Zhang, Guoliang Li, Chengliang Chai.

[VLDB 2022] Learned Cardinality Estimation: Design Space Exploration and Comparative Evaluation
Full Paper, CCF-A Ji Sun*, **Jintao Zhang*** (*equal first contribution*), Zhaoyan Sun, Guoliang Li, Nan Tang.

[Preprint] A Survey of Efficient Attention Methods: Hardware-efficient, Sparse, Compact, and Linear Attention
Full Paper **Jintao Zhang**, Rundong Su, Chunyu Liu, Jia Wei, Ziteng Wang, Pengle Zhang, et al.

[Preprint] Speculative Decoding for Autoregressive Video Generation
Full Paper Yuezhou Hu, **Jintao Zhang*#** (*equal first contribution and corresponding author*).

Research Interests

- My research focuses on **(1)** Efficient Machine Learning System, specializing in accelerating model training and inference; **(2)** Real-time Video Generation, covering the full pipeline from data preparation to model training.

Selected Awards

Tencent Project Up Scholarship	(15 globally/year)	2026
ByteDance Scholarship	(20 globally/year)	2025
Siebel Scholars Scholarship	(80 globally/year)	2023
Zhang Bo Artificial Intelligence Scholarship	(2/year)	2026
Ant Group InTech Scholarship	(10 nationwide/year)	2026